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Development of small vessel advisory and forecast services system for safe navigation and operations at sea

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ABSTRACT

Fishing is the most dangerous profession in the world. To reduce the number of accidents caused by capsizing of vessels, Indian National Centre for Ocean Information Services (INCOIS) has designed and developed an advisory and forecast services system for the Indian ocean regime, which warns users against potential zones and times at which vessel overturning can take place, three days in advance. This paper discusses the development and various aspects of such a system, its decision-making mechanism, and the Standard Operating Procedure (SOP). Warning advisories issued by the system are based upon a newly coined index-Boat Safety Index (BSI). The advisory system is verified with a few real-life incidents of the past. The refinement of the advisory system is planned by updating suitable thresholds for the indices based upon regular feedback from the users. The boats operating along the Indian coast are classified into different categories based upon their beam size, and category-specific advisories are issued.

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Small craft advisory; steepness index; crossing sea index; rapid development of sea index; boat safety index

1. Introduction

India is a peninsular country with a long coastline of approximately 7500 km. As a result of its large vicinity to the sea, various marine activities such as fishing, maritime transport, tourism, offshore exploration, etc. are carried out along the Indian coasts, which contribute immensely to the Indian economy (Ministry of Agriculture & Farmers Welfare 2015). A deep understanding of the climatological, transient, and future state of the ocean, has various practical applications. For example, simulation of historical sea states using numerical models provide information on statistical estimates of the wave parameters for specific regions of interest; whereas the real-time numerical ocean state forecasts are useful for various activities, such as optimal ship routing and safety of vessels, planning of various offshore operations, port and harbour activities, warfare strategies, etc. (Sandhya et al. 2014, 2018; Harikumar et al. 2015; Sirisha et al. 2017). Considering the importance of these forecasts, there have been many attempts in the recent past to improve their accuracy (Deshmukh et al. 2016; Londhe et al. 2016; Swarnali et al. 2018). Prior knowledge of the sea state is important for smooth regulation of marine activities, safe navigation, saving life and livelihood at sea, and national security. Here lies the importance of routine impact-based wave forecasting and its dissemination – to different categories

of users – through various dissemination modes. It is no longer enough to provide a good and timely weather forecast or warning, but there is presently an increased demand for specific information that provides safety to the lives and property of common man (WMO 2015). In simple terms, while there is a realisation of what the weather might be, there is frequently a lack of understanding of what the weather might do. Accordingly, as per WMO (2015) manual, it is urged that all the weather forecasting centres – of its member countries – should convert/upgrade their weather forecast and warning services to a multi-hazard impact-based one, rather than a general one. None of the international wave forecasting agencies, except the National Centers for Environmental Predictions-National Oceanic and Atmospheric Administration (NCEP-NOAA), gives out impact-based weather forecasts that could predict small boat capsizing. NOAA's Small Craft Advisory Services uses predetermined thresholds – that are domain-dependent – for wind speeds and wave heights; this identifies areas of potential risk, but does not relate explicitly to a boat safety index to simplify subsequent decision-making.

In India, Indian National Centre for Ocean Information Services (INCOIS) – an operational forecasting centre under the Ministry of Earth Sciences – is mandated to provide the best possible ocean information

and advisory services to society, industry, government agencies, and the scientific community (Balakrishnan Nair et al. 2013, 2014). Many of these services provide prior information on the sea state well in advance, which includes seas, swells, currents, sea surface temperature, mixed layer depth, etc. (<http://www.incois.gov.in/portal/osf/osf.jsp>). In addition to the regular forecasting services, INCOIS issues alerts and advisories during high waves, cyclones, and other extreme events. For example, The High Wave Alerts (HWA) are issued for a particular region, if there is a chance of prevalence of high wave conditions during the forecast period.

India has over 1500 fish landing centres along with several major and minor ports that significantly contribute to its blue economy. The majority of the vessels operating in Indian nearshore regions are small in size, hence can be termed – although there is no strict classification – as small vessels (Ministry of Agriculture and Farmers Welfare 2015). This category includes fishing boats, merchant boats, passenger boats, service boats; and recreational vessels used for kayaking, yachting, canoeing, etc. There are a considerable number of accidents encountered by small boats (Myrhaug and Kjeldsen 1987). Fishing – with a casualty ratio of 76 per 1000 vessels – for which most of the small vessels are operating in India – is the most dangerous profession in the world (Annual Report, MAIB 2016). Casualties involving small vessels are due to capsizing, foundering, collision, fire/explosion, damage to equipment, etc. with capsizing and foundering as the major contributors (Niclasen et al. 2010). Capsizing often leads to human loss, economic loss, and environmental pollution. The occurrence of these accidents can be drastically reduced by issuing advisories that take into consideration the environmental factors responsible for capsizing (Viggosson 2009) along with proper safety plans accommodating the human factor. The causes of capsizing of ships include synchronous sea state, crossing seas condition, rapid changes in the sea state, extreme weather conditions, human error, flooding, etc. High-resolution models are required to capture such very local events responsible for the capsizing of boats, while most of these events are consequent on the properties of individual waves (Dahle et al. 2005). INCOIS uses phase-averaged third-generation numerical wave models such as Simulating Waves Nearshore (SWAN), WAVEWATCH III (WW3), etc. for wave forecasting purposes (Ris et al. 1994; Tolman 2009). Past studies have shown that the shape of the wave spectra obtained from the phase-averaged models has a relation with the properties of single waves, and hence these models can be used for formulating warning criteria (Dahle et al. 2005). At ESSO-INCOIS,

through the present study, an attempt is made to design, develop and implement a full-fledged Small Vessel Advisory Services (SVAS) system, based on the warning criteria that incorporate factors causing vessel capsizing. The advisory system warns against the possible dangerous future scenarios, and thus reducing vessel overturning probability. We are planning to categorise this forecast and advisory system to, ‘warnings using relevant thresholds agreed with users or practitioners’ type of impact-based forecast and warning services, as specified by the World Meteorological Organisation (WMO 2015). It can only be accomplished by sustained and diligent interactions with the end-users, collection of feedback and incorporating it for the updation of thresholds used in warning criteria of the system along with the inclusion of any extra component if needed. The latest update of the WMO manual (WMO 2018) states and puts forward a guideline that ‘Warning should be provided for the following phenomena, (a) Unusual and hazardous sea-ice conditions and (b) Dangerous sea states’. This particular forecast and advisory service system tries to focus on the second phenomenon, viz. ‘identifying dangerous seas’, and capturing unexpected sea conditions. A large class of society, especially the vulnerable fishing population and those who use small boats for their livelihood or recreational purposes are expected to benefit from SVAS; thereby reducing human, environmental and economic losses associated with vessel capsizing.

The main objective of the paper is to elaborate in detail upon the various aspects of the SVAS system developed and operationalised at INCOIS, which includes its adopted warning criteria, decision-making mechanism, and feedback procedure. The advisory system is semi-operationalised for the east coast of India and will be extended to the entire Indian coast, shortly. Section 2 of the paper briefs about the study region, and the ocean state forecasting system that is used by the SVAS system. Section 3 highlights the principal causes of vessel capsizing, the warning criteria adopted by the system, a single combined index incorporating all these warning criteria, and the Standard Operating Procedure (SOP) adopted by the system. In section 4, some of the sample results from the year-long analysis of the wave model output are discussed along with the standard feedback procedure adopted to improve on the thresholds. The section also explains the boat specific nature of the advisory system and its validation with some real-life examples. The 5th (final) section concludes by summarising components of the system, its warning criteria and preliminary thresholds, and results of validations with real-life cases.

2. Study area and modeling system

In its first phase, SVAS is set-up at ESSO-INCOIS on a semi-operational basis for the Bay of Bengal region that extends from 4° N to 22.5° N latitude and 77° E to 90.5° E longitude (Figure 1). It is evident from the subscribed user's data at ESSO-INCOIS that the areas nearer to the coast have a higher density of small vessels, and thus are very important concerning the advisory system. The advisory and forecast services system takes into consideration certain warning indices, that are calculated using wave parameters obtained from operational numerical wave model forecasts. The system uses an unstructured SWAN modelling set-up for the East Coast of India that is presently semi-operational at ESSO-INCOIS issuing wave forecasts 3 days in advance at 3-hourly intervals (Sandhya et al. 2018) and is forced by 25 km × 25 km winds from the European Centre for Medium-Range Weather Forecasts (ECMWF) at 3-hourly intervals along with the usage of General Bathymetric Chart of the Oceans (GEBCO) 30 arc second bathymetry. The computational grid of this system has the highest (500 m) resolution in the coastal areas, which relaxes to 50 km as we move offshore. The model has been validated using observations obtained from satellite as well as wave rider buoys, and the performance is analyzed comprehensively (Sandhya et al. 2018). Figure 1 shows the region for which advisory is issued on a semi-operational basis. The highest resolution

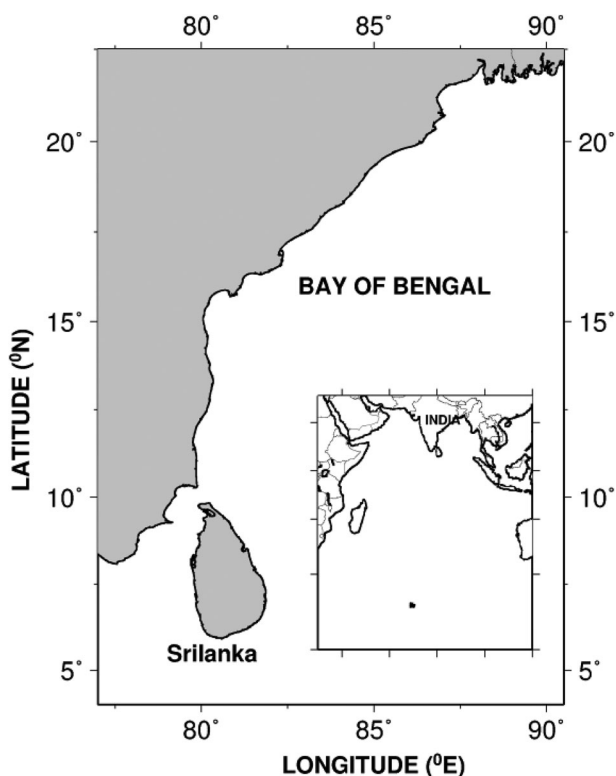


Figure 1. Study Area, and the model domain (given in inset).

of the modelling system in the coastal areas helps in determining and identifying the areas prone to the boat capsizing more accurately and with minimum computational capacity.

3. Methodology

3.1 Causes of capsizing of small vessels

The capsizing of vessels causes human, environmental, and economic loss, and there are various factors responsible for it. Most of the accidents occur not by a single, but a series of causal factors acting together (Niclasen et al. 2010). Although several causes of capsizing are interrelated and occur simultaneously, for classification purposes they are divided into three major categories, viz, design and stability of the vessel, human error, and environmental hazards.

Small vessels are the most diverse class of vessels in the sea with a huge variety in their sizes, shapes, hull sizes, etc.; and they are not properly represented by stability evaluation methods (Womack 2003). Mostly in designing the stability of such vessels, static effects are only taken into account while ignoring the dynamic effects (Senjanovic et al. 1997). Inadequate stability analysis of a boat in its design stage results in an increase in the vulnerability of vessels for capsizing (Dahle and Myrhaug 1995). Human error is responsible for 75–96% of vessel casualties directly or indirectly and is caused mainly because of an incorrect decision; an improperly performed action; or lack of action due to fatigue, inadequate communications, insufficient technical knowledge, etc. (Rothblum 2000). Environmental factors such as single individual waves play an important role in boat capsizing, with some higher waves having the potential to even capsize large ships (Olagnon and Iseghem 2000). The wind heeling phenomenon – high magnitude lateral force exerted on the boat by the wind blowing over the seas, causing it to lean on one side – often compromise the structural integrity and stability of the vessel, thereby increasing its capsizing probability (Deakin and Wright 2005). Wave directionality and steepness have adverse impacts on vessel stability (Womack and Johnson 2004; Ayaz et al. 2006). In ‘synchronous wave conditions’ – where the period of the vessel is the same as that of the wave – the resonance between the boat and the wave leads to excessive rolling, thereby increasing the chances of water entering inside the vessel (Dahle and Myrhaug 1995; Dahle et al. 2005).

3.2 Warning criteria adopted for SVAS

The efficient working of the SVAS system depends on the appropriate choice of warning criteria such that

the majority of causal factors for capsizing are taken into consideration along with their severity. There are several studies undertaken in the past that take into consideration various aspects of extreme single individual waves, their probabilities of occurrence, distribution, etc. (Myrhaug and Kjeldsen 1984, 1987; Brodtkorb 2004; Kjeldsen 2004). At INCOIS, the advisory for small vessels against capsizing is issued based on the following warning criteria.

3.2.1. Steepness index

High waves in breaking conditions adversely affect the stability of vessels, with accidents due to waves not only caused by the very highest waves (Myrhaug and Kjeldsen 1984; Kjeldsen 2004). Also, the breaking wave height equal to the beam of vessels is sufficient to capsize them (Maritime and Coastguard Agency 2004). To find the contribution of significant wave height and wave breaking in the capsizing, joint probability studies of both these parameters were carried out by several researchers, ultimately stating their interrelation in severe sea states (Myrhaug and Kjeldsen 1984; Brodtkorb 2004). These studies conclude that steep waves with high – not just the highest – significant wave heights are responsible for boat capsizing (Gunson et al. 2001), and thus imperative to consider a warning criterion that takes into consideration both the parameters i.e. steepness and significant wave height. Such a warning criterion is proposed in Work Package 8 (WP8) of MAXWAVE project (Savina et al. 2003). This warning criterion (Savina et al. 2003; Rosenthal and Lehner 2008) is approved by The Joint Commission for Oceanography and Marine Meteorology (JCOMM) and can be represented mathematically as

$$I_{\text{steepness}} = \frac{S_s}{0.05} \times \frac{H_s}{h_0} \quad (1)$$

where $h_0 = 2.5$ m (assumed for Indian seas)

The index which takes into consideration the wave steepness (S_s) and significant wave height (H_s), and is termed as *Steepness Index* ($I_{\text{steepness}}$). The mathematical representation of $I_{\text{steepness}}$ contains 0.05 in the denominator, which stands for ‘PM-Limiting Zero Crossing Steepness’, whereas h_0 is a constant value that is region-specific. The threshold of the index varies with the value of h_0 , but is currently fixed at 2.5 m for the Indian seas based on our experience in wave forecasting.

3.2.2. Crossing sea index

Wave directionality with respect to a vessel plays an important role in determining its stability (McTaggart

et al. 2002). For example, in the ‘Following sea condition’ – where the direction of the wave and vessel are the same – the probability of boat capsizing increases due to prolonged encounter between the vessel and the steep wave; in the ‘Head sea condition’ – where the direction of the vessel is opposite to that of the wave – provides maximum stability, but with huge slamming forces on the vessel that may cause shifting of its cargo; in the ‘Beam sea condition’ – where the direction of vessel and wave are perpendicular to each other – often large rolling angles of vessels occur that may result into boat capsizing during extreme conditions (Al Salem et al. 2006). The experience of sailor plays a very important role during such multidirectional wave encounters – where two or more wave trains approach the vessel from different directions – termed as ‘Crossing Seas’. Such a sea state has a high risk of boat capsizing and it is found to be a contributing factor in one out of four boat accidents (Toffoli et al. 2005). The Directional Spread parameter obtained from phase averaged wave models estimates the extent of crossing seas by determining the range of directions that contain wave energies. The value of directional spread ranges from zero, for unidirectional spectrum; to $\sqrt{2}$, for uniform spectrum. Inclusion of the directional spread along with the severity of sea states – often denoted by significant wave height – in warning criteria is imperative. One such warning criterion that includes both of them is termed as Crossing Sea Index ($I_{\text{crossing seas}}$) and is proposed in the past (Savina et al. 2003). It is mathematically represented as

$$I_{\text{crossing seas}} = \frac{1}{2} \times H_s \times \exp(-10 \times (\sigma_s - 1)^2) \quad (2)$$

The index is calculated from the Significant Wave Height (H_s) and the Directional Spread (σ_s) with its threshold value determined by analyzing the wave model outputs. The threshold determination procedure will be discussed in detail in the succeeding sections.

3.2.3. Rapid development of sea index

It was inferred from earlier studies that four out of five accidents happened when the region experienced rapid changes – where there was an increase in the wind component of sea state by more than 20% within 6-hour duration – in the sea state. It was also concluded that rapid changes in sea state result in less reaction time for the sailors, thereby causing accidents due to a series of human errors (Toffoli et al. 2005). The ‘rapid development of sea’ criteria or the ‘enhancement factor’ (Z_{6h}) is a measure of the change in the wind

component of the generic wave parameter in 6 h. If the parameter value at the initial time is (ξ_i) and that after 6 h is (ξ_f), then it can be mathematically represented as

$$Z_{6h} = \frac{|\xi_i - \xi_f|}{|\xi_i|} \quad (3)$$

The wind sea wave height (H_{sea}) obtained from the wave model represents the wind component of the sea state in the most suited way. As a result, the generic wave parameter mentioned in the above equation can be replaced by wind sea wave height (H_{sea}), and so the revised equation is

$$Z_{6h} = \frac{|H_{sea_i} - H_{sea_f}|}{H_{sea_i}} \quad (4)$$

If $Z_{6h} \geq 0.2$, then a warning is to be issued (5)

As suggested in the literature, any value of the enhancement factor above 0.2 is considered as dangerous.

3.2.4. Boat safety index

It is to be noted that warnings – in the textual as well as image formats – are issued to the users when any of these indices surpass their threshold value. To facilitate the information interpretation for the forecaster, and avoid confusion arising out from multiple images – separate for each warning criteria – a single index, namely Boat Safety Index (BSI) is proposed that combines the effect of all the indices and represented mathematically as

$$BSI = S_{steepness} + S_{crosssea} + S_{rapiddev} \quad (6)$$

In Equation (6), BSI stands for Boat Safety Index; with $S_{steepness}$, $S_{crosssea}$ and $S_{rapiddev}$ denoting intermediate indices for steepness, crossing seas, and rapid development of sea respectively. These intermediate indices are obtained from the fundamental indices through mathematical relationships represented as

$$S_{steepness} = 1, \text{ if } I_{steepness} > \text{threshold, else } S_{steepness} = 0 \quad (7)$$

$$S_{crosssea} = 2, \text{ if } I_{crossing\ sea} > \text{threshold, else } S_{crosssea} = 0 \quad (8)$$

$$S_{rapiddev} = 4, \text{ if } Z_{6h} > \text{threshold, else } S_{rapiddev} = 0 \quad (9)$$

The value of BSI ranges from 0 ($BSI = 0 + 0 + 0$) to 7 ($BSI = 1 + 2 + 4$), where the values 1, 2 and 4 are selected to represent the surplusses of indices $S_{steepness}$, $S_{crosssea}$ and $S_{rapiddev}$ respectively, that helps in understanding the prevalent contribution of each of these indices to the BSI. Such a value assignment is mandatory to analyze the issued advisory in conjunction with the feedback from the users, and this is explained in Section

4.2. If $BSI = 0$, then none of the thresholds corresponding to the three warning criteria have been surpassed and the sea state could be considered safe. For $BSI = 7$, all the thresholds have been crossed; hence, it is the most dangerous condition for the seas. It is to be noted that for any non zero value of BSI, any one of the indices or more has been surpassed, and so can be considered as dangerous.

Figure 2 contains the spatial representation of BSI at 0830 IST (UTC + 0530 hrs) on 13 January 2014, which is based on the wave parameters obtained from unstructured SWAN model simulations. The BSI in the figure ranges between 0 and 3. The region with $BSI = 0$, – i.e. the region with a dark blue patch – is safe concerning boat capsizing; whereas the regions with $BSI = 1$ and 2 are unsafe, as they have surpassed their thresholds for steepness index and crossing seas index respectively. On the other hand, the region with $BSI = 3$ has both the thresholds for steepness and crossing sea indices surpassed. It can also be seen that the rapid development of sea index has not exceeded its threshold anywhere for the given time step, as the region with $BSI = 4$ is absent.

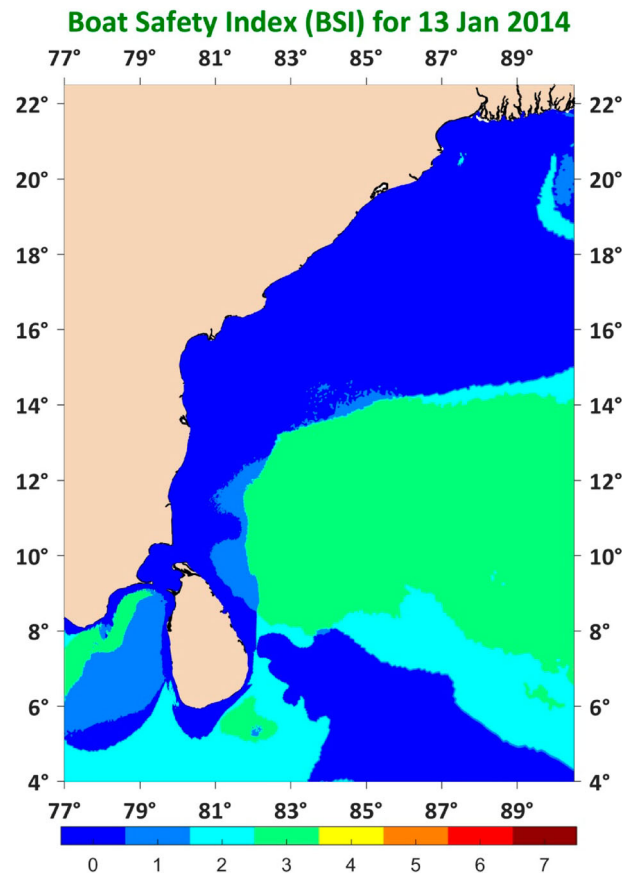


Figure 2. Spatial plot of Boat Safety Index (BSI) for 0830 IST on 13 January 2014.

3.3. Decision-making mechanism of SVAS

The advisory and forecast services system discussed here uses averaged wave parameters from the unstructured SWAN model at each computational grid point to issue warnings and/or alerts related to boat capsizing. The entire process involves a decision-making mechanism that is represented pictorially in Figure 3 for a single computational grid point, and it can be replicated at each of the model computational grid points.

In the process of decision making for a grid point, firstly, all the three indices – i.e. ‘steepness index’, ‘crossing sea index’ and the ‘rapid development of sea index’ – are calculated at each grid point using Equations (1), (2) and (4) respectively, with the help of the wave parameters obtained from the SWAN model. Secondly, these indices are checked for

surpassing their respective thresholds; and their ‘intermediate indices’ are calculated using Equations (7–9). Thirdly, these ‘intermediate indices’ are used to calculate BSI by using Equations (6), that help in identifying the warning and/or alert zones. Fourthly, the values of the indices as well as their respective images for the entire domain are stored in a database, which can be used by the designed feedback mechanism for the threshold updation purposes. Lastly, the advisory system is made boat-specific with the help of H_s is over the warning and/or alert regions.

The warnings and/or alerts are disseminated in the local language are issued directly to the end-users, boat owners, and collaborating NGOs through various modes; for example, INCOIS website (www.incois.gov.in), mobile phone SMS, voice calls, FM radio, TV, mobile applications, social media, etc.

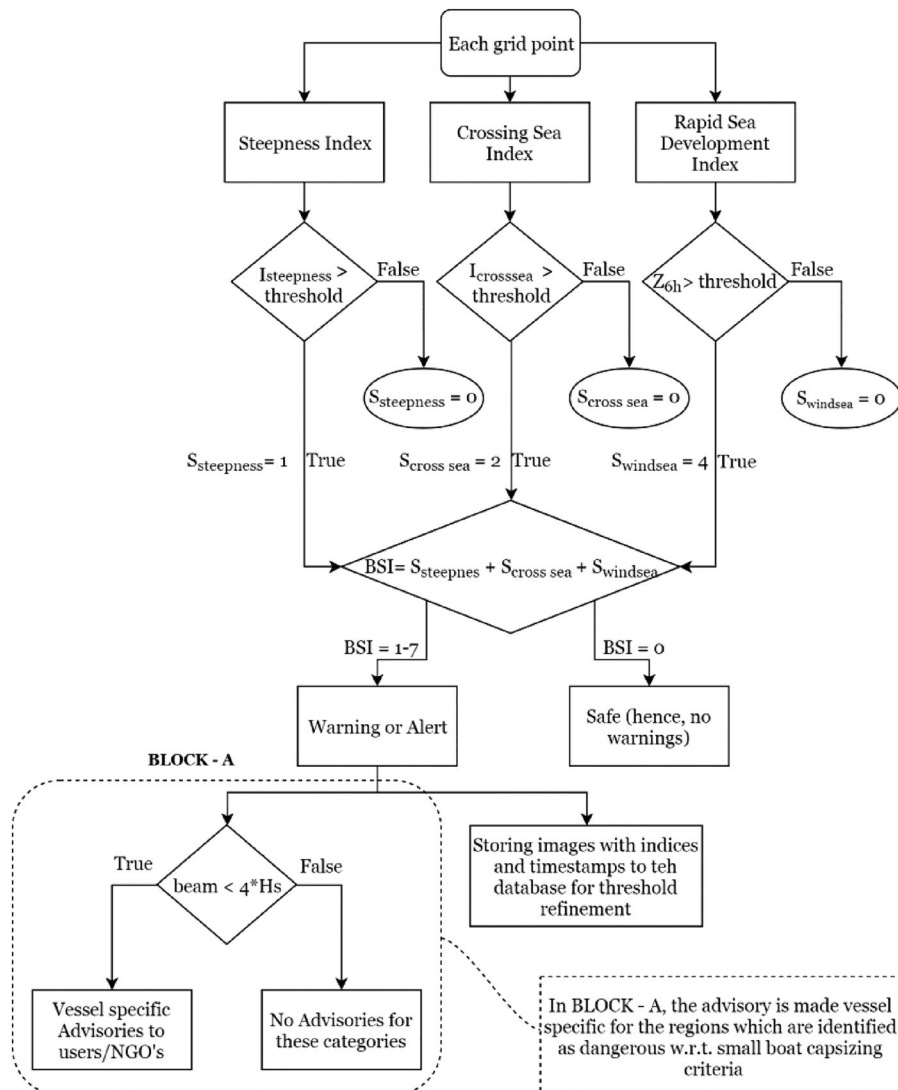


Figure 3. Flowchart of decision-making mechanism for SVAS.

4. Results and discussions

4.1. Determining thresholds for the warning indices

SVAS uses three types of warning criteria (or indices) namely steepness index, crossing sea index, and the rapid development of sea index or enhancement factor. It is important to determine the thresholds of these indices and warn the end-users if any of the thresholds have exceeded.

As already mentioned in section 3b, the threshold for the enhancement factor is kept at 0.2, which means that the areas where the enhancement factor is equal to or greater than 0.2 are identified as potentially dangerous for small craft and a warning is issued. There are no definite thresholds mentioned in the literature for the other two indices, and hence they are estimated by analysing wave model outputs for an experimental period. Insight from user feedback is used to update these thresholds over a period of time, and this approach is discussed comprehensively in later sections.

Presently, the high wave alerts/warnings issued by ESSO-INCOIS for several regions help the user community in understanding the severity of sea state well in advance and make informed decisions. If the predicted significant wave height (swell height) lies between 3 m (2.5 m) and 3.5 m (3 m), the coastal regions are put on a necessary alert; and if it is > 3.5 m (3 m), then high wave warnings are issued. These alerts are generalised and can be used by a variety of users like fisher folks, Indian coast guard, offshore firms, engineering consultants, port and harbour authorities, coastal residents, etc.; whereas SVAS focuses specifically on the factors related to the capsizing of small vessels, and so are important only to these users. A detailed analysis of wave model outputs (hindcast) was performed to decide

upon the preliminary threshold values of the steepness index and crossing sea index. In the beginning, it was assumed that the time during which high wave alert was issued for a particular region might also have been dangerous for small boats in general, i.e. either of the indices (steepness/crossing sea) or both the indices might have exceeded their threshold values. Under this assumption, studies were carried out for the time instants and regions where high wave alerts were issued, and their correspondence with respective steepness index and crossing sea index plots was established.

The spatial patterns of significant wave height (panel (a) in Figures 4–7), steepness index (panel (b) in Figures 4–7), and crossing sea index (panel (c) in Figures 4–7) were plotted for high wave alert conditions in 2014, with each for all four dominant seasons in the Indian Ocean. It may be noted that the colour-bars of steepness index and crossing sea index plots have finer intervals in the ranges near to their thresholds, to capture minor changes. The discretization is finer at ranges, where the probability of occurrence of the thresholds is more (steepness index values and crossing sea index values close to 0.6 and 0.4 respectively), and coarser elsewhere. Seasonal analysis for qualitatively determining the thresholds is illustrated below.

4.1.1. Winter season (January–February)

Figure 4(a–c) represent the significant wave height, steepness index, and crossing sea index respectively, obtained from the outputs of the modelling system for 0830 IST on 13 January 2014. It is known that the waves with significant wave height above 3 m can be considered as dangerous for small crafts. However, it is apparent that if these high waves would have existed along with steep waves and also with a crossing sea condition, the situation

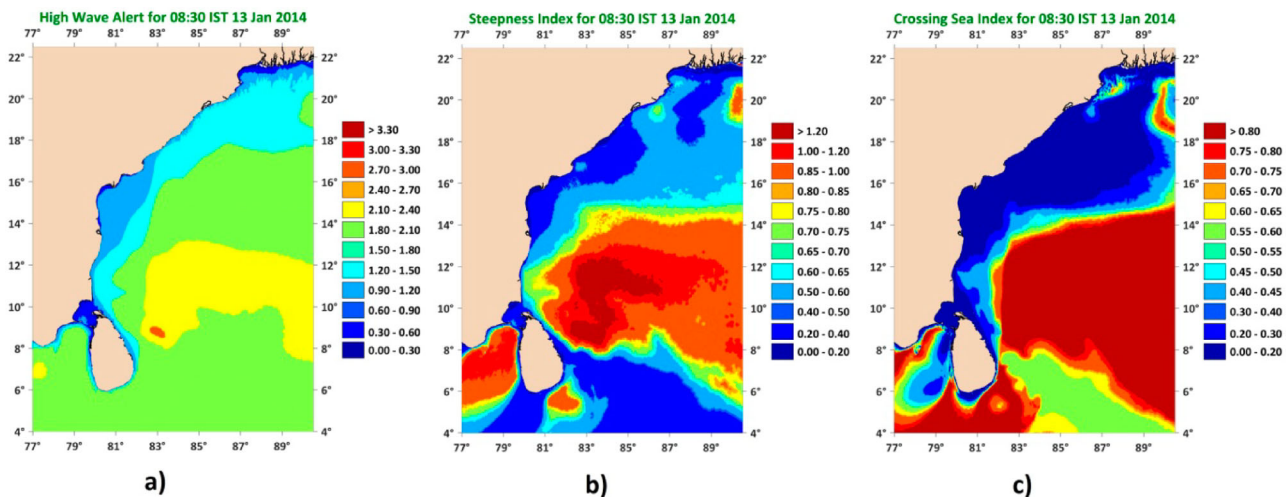


Figure 4. Spatial plots of significant wave height, steepness index and crossing sea index respectively for 0830 IST on 13 January 2014.

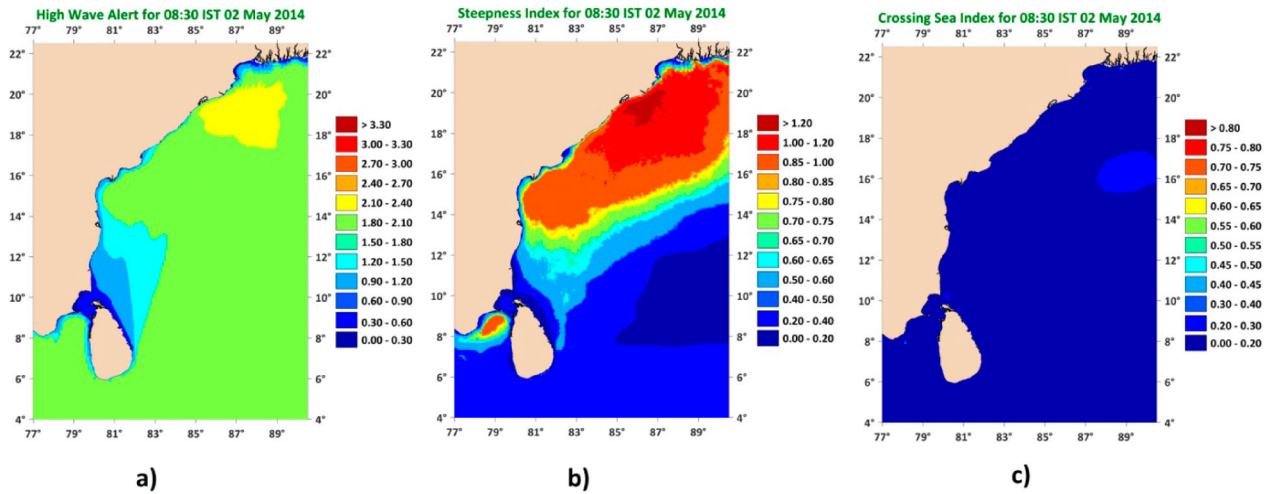


Figure 5. Spatial plots of significant wave height, steepness index and crossing sea index respectively for 0830 IST on 2 May 2014.

would have been highly dangerous even when the wave heights are less than 3 m. Hence, in the analysis, we started cross-checking the presence of this condition for waves with a significant wave height of 2 m or above. Accordingly, the patch in Figure 4(a) that corresponds to the range between 2.1 and 2.4 m significant wave heights can be considered as dangerous, and obviously, the patches for higher values are also perilous. In the steepness index plot i.e. Figure 4(b), the patch corresponding to the dangerous patch in Figure 4(a) lies in the range or just near to 0.8–0.85, and this range may be suitable for fixing the threshold. Similarly, if we compare crossing sea index in Figure 4(c) with Figure 4(a) we can observe that the range 0.65–0.7 is complying well with the patch of 2.1–2.4 m, and may be taken as a potential threshold for the crossing seas index.

4.1.2. Pre-Monsoon season (March–May)

Figure 5(a–c) depict significant wave height, steepness index, and crossing sea index respectively for the pre-monsoon season for 0830 IST on 2 May 2014. Even in this case, the range of steepness index which corresponds to the wave height in between 2.1 and 2.4 m is 0.8–0.85 which further strengthens the assumption of keeping 0.8–0.85 as its threshold range. As far as the crossing sea index is concerned, there is no visible pattern in the whole model domain. The low value of the index at this instant maybe because of the weak interaction of different wave trains in the model domain and consequently, the figure does not give any indication of a probable threshold for crossing seas.

4.1.3. Southwest monsoon season (June–September)

In the southwest monsoon season, the sea is generally rough and it is expected to have high values of

significant wave height most of the time. Figure 6(a–c) are snapshots for 0830 IST on 20 June 2014; and show the spatial variations in significant wave height, steepness index, and crossing sea index respectively. The increase in wave height in the whole model domain is evident in Figure 6(a). Most of the areas in Figure 6(b) show very high values of steepness index, but by thorough scrutiny, it is evident that the range 0.8–0.85 again coincides with that of significant wave height range 2.1–2.4 m, further reinforcing our assumption of the threshold for the index. In Figure 6(c), there are patches of a high cross-sea index in the region corresponding to dangerous significant wave heights (2.1–2.4 m). These patches of crossing sea index are in the range of 0.65–0.7 for high wave regions, thus re-iterating the assumption of having the 0.65–0.7 range as the threshold for crossing sea index.

4.1.4. Post-monsoon season (October–December)

Figure 7(a–c) depict significant wave height, steepness index, and crossing sea index respectively for 0830 IST on 5 November 2014. It can be seen that for steepness index and crossing sea index, the patch with the range in between 0.8–0.85 and 0.65–0.7 respectively, correspond to a significant wave height range of 2.1–2.4 m, and are similar to the results obtained from the analysis of other seasons.

By interpreting the analysis results for all the seasons, it is evident that the ranges of 0.8–0.85 for steepness index and 0.65–0.7 for crossing seas index coincide very well with the high significant wave heights. Hence, the preliminary thresholds are kept at 0.8 for the steepness index and 0.65 for crossing sea index. The lower limits are taken to avoid any misses in

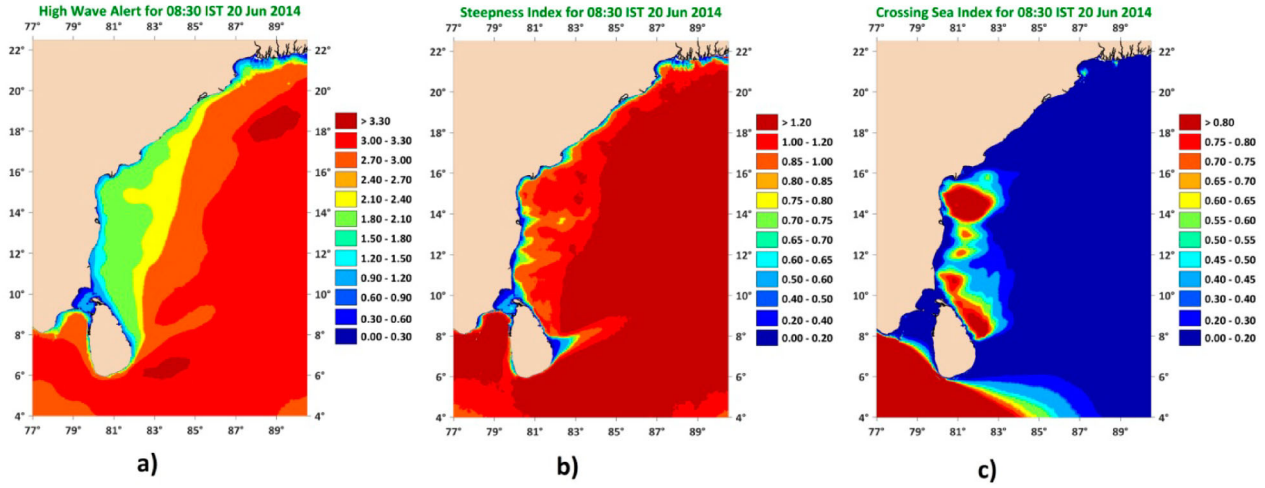


Figure 6. Spatial plots of significant wave height, steepness index and crossing sea index respectively for 0830 IST on 20 June 2014.

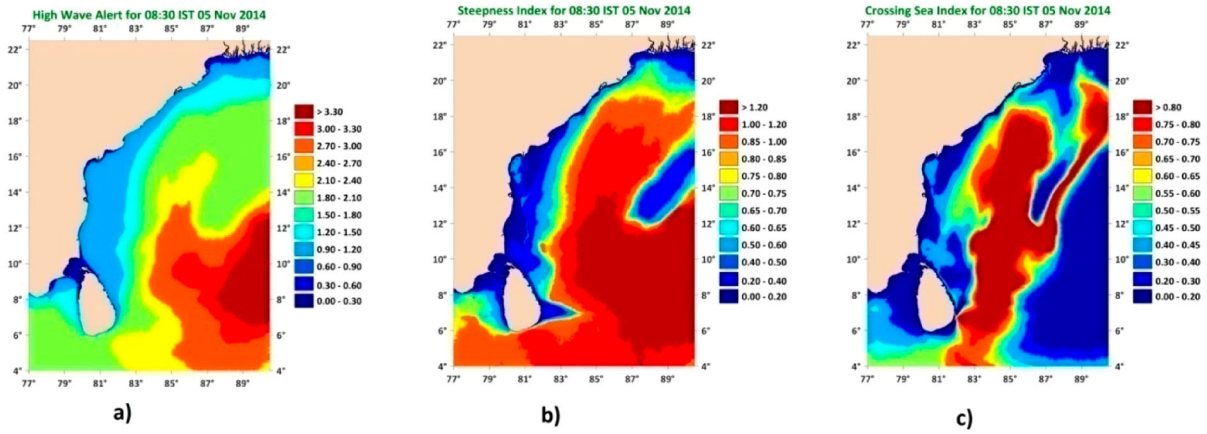


Figure 7. Spatial plots of significant wave height, steepness index and crossing sea index respectively for 8.30 IST on 5 November 2014.

warnings and/or alerts issued by SVAS. Thus,

$$\text{Threshold for } I_{\text{steepness}} = 0.8,$$

$$\text{Threshold for } I_{\text{crossing seas}} = 0.65$$

4.2. Small vessel advisory with text and images

In the earlier sections, we have discussed the procedure to calculate BSI for each time step depending on the wave model forecasts. It is also clear that depending on the BSI value we can identify the regions which are safe and otherwise, for small vessel capsizing. As the model output is available on a 3 hourly basis, we have, in all, model forecasts for 8-time steps in a single day; and hence, BSI can be calculated at all these time steps. The advisory system categorises a region as safe, alert, or warning zone. The process of segregation can

be explained with a sample day's BSI output, which is represented in Figure 8; with the blue colour for areas with BSI = 0, and orange colour for non-zero values of BSI.

Figure 8 shows the spatial pattern of zero and non-zero values of BSI for the region considered at 3-hourly intervals for a day (2 May 2014); each subplot represents a single time step of the day. The advisory is specific to a single day which is issued once all the 8-time steps available for a single day are analyzed. The regions for which BSI is non-zero for at least 7 out of 8-time steps in a day, 'WARNING' is to be issued. The regions where BSI is non-zero at any time step from 5th or later onward (up to 8th time step), ALERT is to be issued. It is to be noted that the regions which are included in the ALERT category are not at all included in the WARNING category. The areas which are not included in either WARNING or ALERT are marked as SAFE.

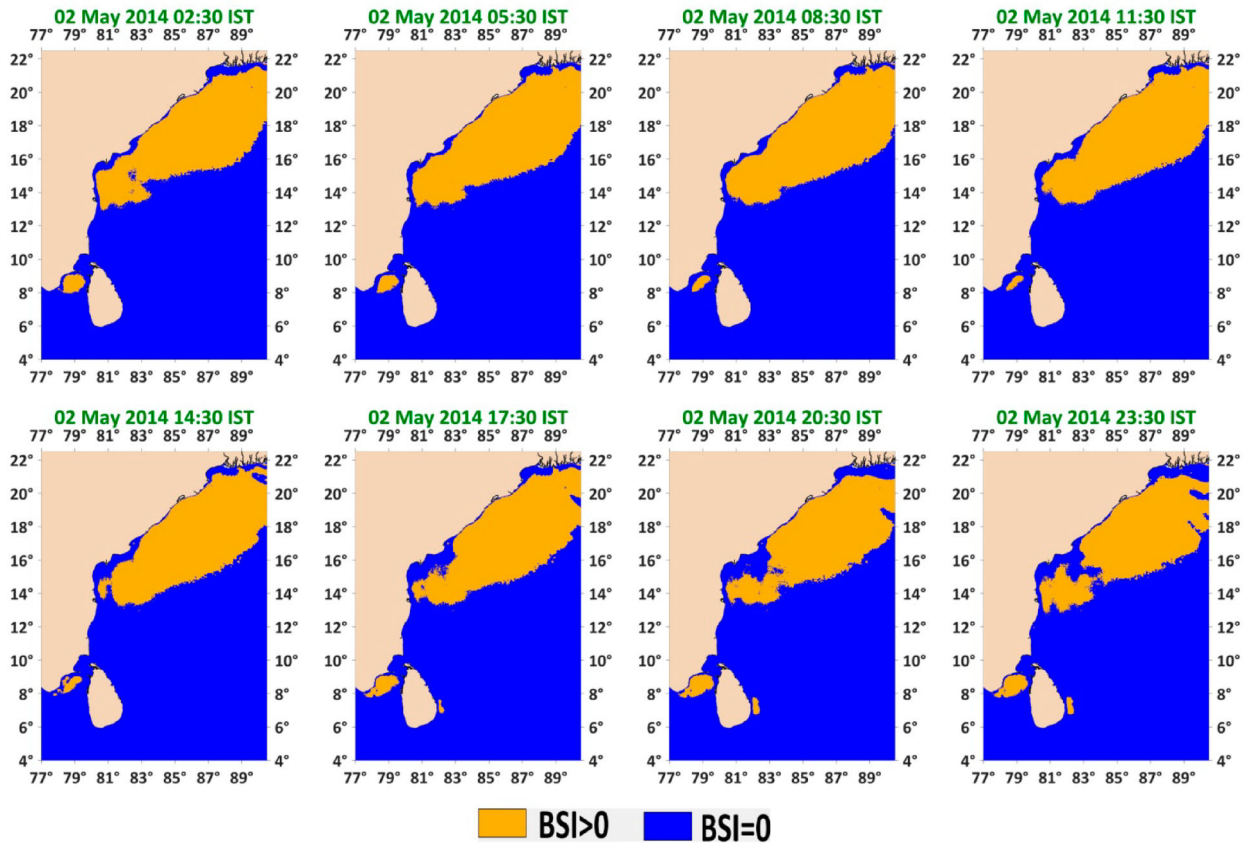


Figure 8. The spatial pattern of BSI for a single day at 3-hourly time intervals.

The criterion applied to decide whether it is a warning, alert, or safe is also supplied with the advisory to get the users/NGOs perceived on the concept of delineating the threat as warning or alert. A typical representative picture of the final advisory service is shown in Figure 9. The regions in red colour are those where WARNING is to be issued on 2 May 2014. Similarly, ALERT is signified by Orange and SAFE by Blue.

4.3. Updating thresholds by a feedback mechanism

ESSO-INCOIS has various methods by which it collects feedback from users for most of its products. Prominent among them are Non-Governmental Organizations (NGOs) such as M S Swaminathan Research Foundation (MSSRF), Pondicherry Multipurpose Social Service Society (PMSSS), IFFCO Kisan Sanchar Ltd. (IKSL), coastal institutes and universities, Boat Owners Association, coast guard, etc. which work on the ground level and help the users in deciphering the ocean state forecast and information services rendered from ESSO-INCOIS.

To make use of this feedback for the improvement of the system, the forecasted images as well as the

Small Craft Advisory for 02-May-2014

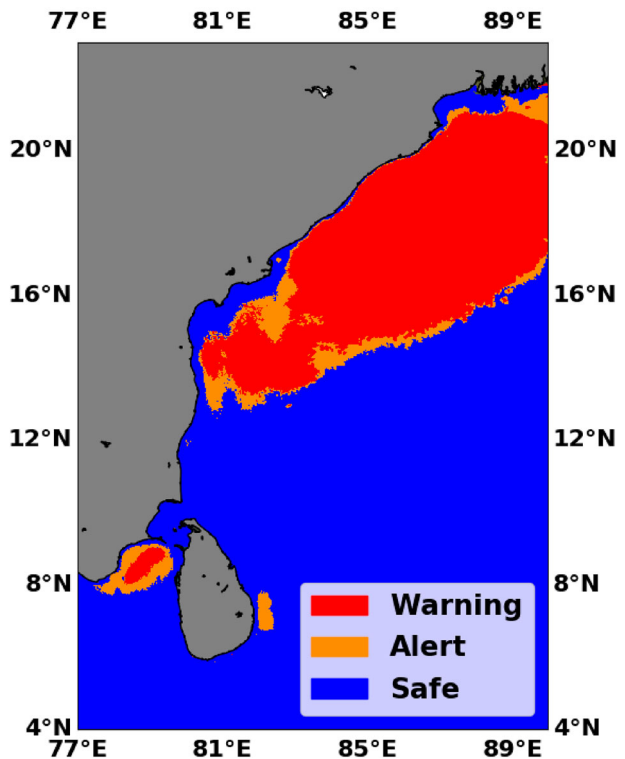


Figure 9. SVAS spatial plot for 2 May 2014.

respective indices – for the instants when the warnings were issued – are stored in a database. Once a false warning is detected, concerned inputs from the proposed feedback mechanism are analysed; the particular index which was responsible for the false warning is identified and its threshold is updated accordingly. This is done by carefully examining BSI of these regimes, and using Table 1 as explained in the next paragraph.

Table 1 shows how the indices, that exceeded their thresholds, can be determined from the BSI at a particular region. For example, if BSI = 3, we can easily reckon that both crossing seas as well as steepness index, were exceeded. Similar bifurcation can be carried out for BSI ranging from 0 to 7, where BSI = 0 is the safest condition for a boat to venture. Thus the strong feedback system helps in detecting the false alarms and the misses along with the regime details. Over a while, if a particular threshold is surpassed during each false alarm condition or miss condition, then the value of that particular threshold is updated, and this process goes on until the faulty alarms/misses are minimum. It is believed that in due course of time the thresholds will be refined and a robust advisory service system will be available for the Indian Ocean region, thus providing safety to small vessels and their owners.

4.4. Issuing boat-specific advisories

India has a wide variety of small boats operating throughout its long coastline, most of which operate within 100 km of the coast. The types and specifications of these boats differ based on the purpose for which it is used, and also the state boundary under which it operates. Edwin et al. (2014) mention some details such as the overall length, beam size, depth and power of the crafts for vessel types such as Hook and Line, Gill Netters, Purse Seiners, Trawls and Dol Net that are used in each of the Indian coastal states for mechanised marine fishing. It is noted that for the small boats used in the Indian coastal waters, the beam length varies from 2 to 6 m, their overall length ranges from 8 to 24 m, and depth varies between 1.5 and 5.25 m (Edwin et al. 2014).

Table 1. Indices exceeded with different values of BSI.

BSI	Indices which exceeded their threshold
0	None
1	Steepness index
2	Crossing sea index
3	Steepness index + Crossing sea index
4	Rapid dev. index
5	Rapid dev. index + Steepness index
6	Rapid dev. index + Crossing sea index
7	Rapid dev. index + Steepness index + Crossing sea index

After analyzing the boat accident data, Maritime and Coastguard Agency (2004) stated through accident analysis that beam length and wavelength at a particular location have a direct correlation with the phenomenon of the boat capsizing. Specifically, for any given wave height, vessels below a particular beam length are more prone to capsizing. Mathematically speaking,

$$\text{Beam} < 4 \times H_s \quad (10)$$

is the condition for the capsizing of vessels. Hence, the following criterion is used in addition to BSI for making the advisory service boat-specific. The regions having non-zero BSI have a high probability of small boat capsizing. For the regions with non-zero BSI, the beam length of vessels below which the boat has a high probability of capsizing is calculated by Equations (10) using the H_s obtained from the wave forecasting system. The boats with beam length less than this threshold are prone to capsizing; and hence, an advisory is issued only to vessels in this category, making the advisory system specific to the size of the vessel.

4.5. Verification of the SVAS system configuration using real-life examples

To establish the advisory service system for small vessels, it is very important to validate the proposed mechanism with some real-life incidents to establish its adequacy and reliability. At the same time, obtaining reliable accident data for small vessels is a challenging task as the incidents of vessel capsizing may not be recorded or communicated systematically. However, INCOIS has extensive collaboration with the Indian Coast Guard and several NGO partners through whom we collect information regarding such incidents. In this section we demonstrate how BSI is used to identify high problematic areas with respect to small vessel capsizing, using some real-life cases that occurred in the domain of the present study.

Figure 10 shows the spatial plots of BSI over the entire domain for the timesteps nearest to those of the accidents. Figure 10(a–c) correspond to the accidents occurred in the southern areas of the domain i.e. off Tamil Nadu (TN) coast, whereas Figure 10(d) corresponds to the accident which occurred at the northern part of the domain i.e. off West Bengal (WB) coast. It can be seen from Figure 10(a) that darker patches have appeared in the region off TN coast where BSI > 0 on the date and time of the accident i.e. 23 April 2019, 14.30 hrs. The pattern has been more or less the same for the nearby timesteps. As per the advisory BSI > 0 are the regions where there is a probability of capsizing

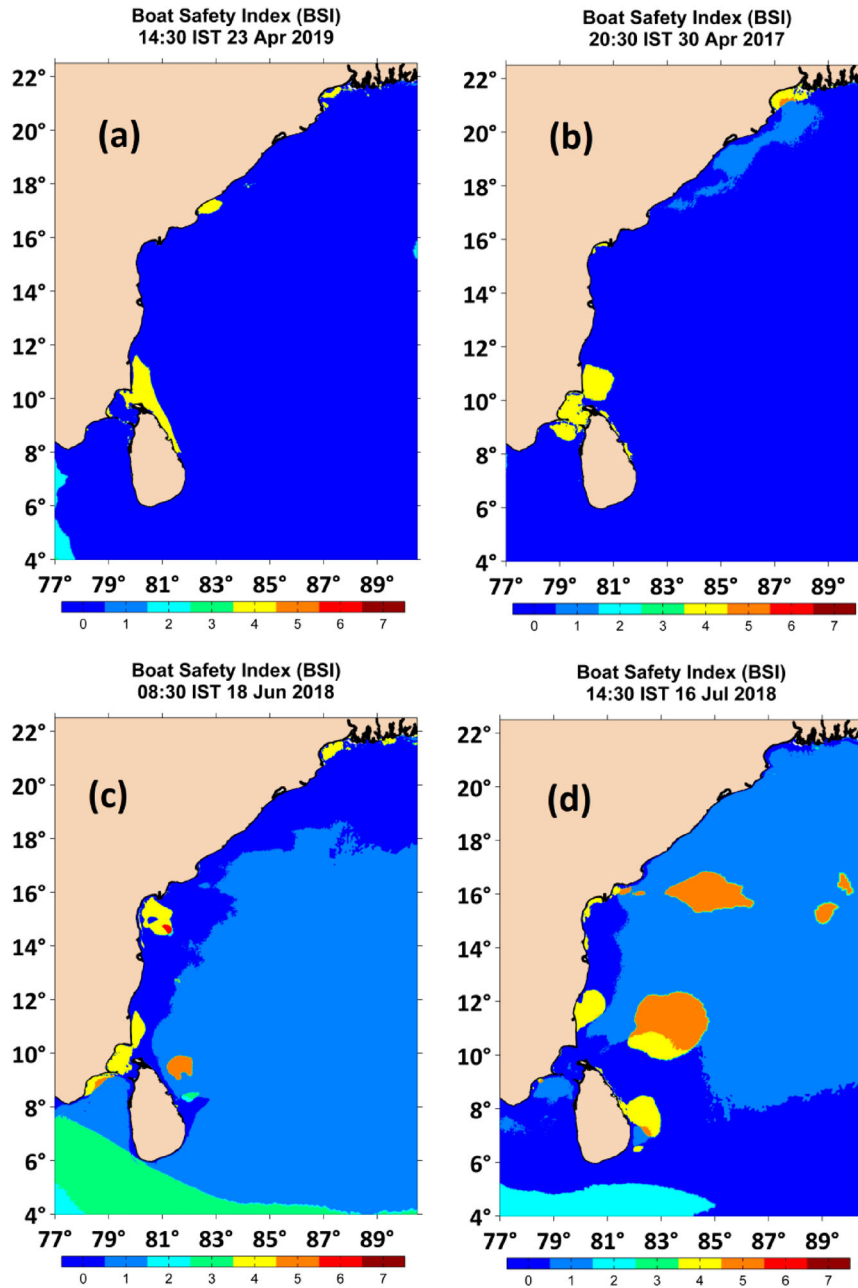


Figure 10. The spatial plot of Boat Safety Index (BSI) for (a) 1430 hrs IST 23 April 2019 (b) 2030 hrs IST 30 April 2017 (c) 0830 IST 18 June 2018 (d) 1430 IST 16 July 2018.

and the above observation complies with the predicted scenarios, thus validating the mechanism adopted and the modelling system used. To make the advisories boat-specific, the warning can be issued with the region of danger in conjunction with the category/size of boat in an inclusive manner, by using the significant wave height (Hs) information from the model in equation (10). In this case, the Hs at the concerned region was up to 0.9 m, and so by using equation (10), it can be warned that the regions with $BSI > 0$ are dangerous for all the boats with beam size less than 3.6 m. Similar

conclusions can be drawn for the other two capsizing incidents along the TN coast (Figure 10(b and c)) that occurred on 30 April 2017, 20.00 hrs, and 18 June 2018, 08.30 hrs respectively. The $BSI > 0$ zones off Chennai in the above two cases have Hs of the orders of 1.5 m and 2.1 m for the corresponding time steps, which translates into a warning/alert advisory to be issued for boats with beam length less than or equal to 6 m and 8.4 m respectively (derived using equation 10). Data corresponding to the nearest time steps from the model output is used here for the validations. Figure

10(d) corresponds to a series of 5 or more accidents which occurred off the coast of WB on 16 July 2018 around 14.30 hrs. It can be seen from the figure that for this case majority of the region is darker with $BSI > 0$ and can be considered as dangerous. The series of accidents which occurred off Kolkota on 16 July 2018 certifies our hypothesis that non-zero BSI signifies dangerous regions. The H_s corresponding to the unsafe regions off the WB coast is around 3.3 m, which implies that advisories have to be issued for all the boats having beam size less than or equal to 13.2 m (by applying equation 10). This beam length range covers almost every boat operating in the Indian region and hence this is one of the most dangerous cases from our model output analysis. It was encouraging to see that the places and time windows of boat accidents on 16 July 2018 off Kolkota coincided with our results of boat-specific SVAS advisories. Hence this case especially justifies the use of BSI in identifying the dangerous zones, by showing a large area of non-zero BSI off the northern coast, where the series of boat accidents occurred. Thus all the 4 cases analyzed showed good agreement with our advisories, which reiterates the usefulness and reliability of the SVAS system.

5. Conclusions

The present study elaborates on the development of a Small Vessel Advisory and forecast Services System for the eastern coasts of Indian Ocean initiated at ESSO-INCOIS to facilitate safe navigation of small vessels, to warn them against capsizing, and for better operational planning. It takes into consideration the factors responsible for boat capsizing, which can be categorised broadly into the design and stability aspects of these vessels, environmental factors, and human errors contributing to the overturning of boats. There is difficulty in taking into consideration all of these factors due to the dearth and inadequate details of data on small vessels operating in the Indian waters. Several studies were carried out in the past to find the correlation between the factors responsible for boat capsizing and the parameters obtained from the wave models, and accordingly, various warning criteria were proposed. Of these, the warning criteria which take into consideration major causes of capsizing, apart from being operationally feasible were selected in this work. Advisories are issued based on the information obtained from applying these warning criteria on the wave model outputs.

The three warning criteria selected are the steepness index, crossing sea index, and rapid development of sea index. The steepness index takes into consideration the

effects of wave breaking along with the severity of sea state, crossing sea index takes care of multi-directional waves by considering the interactions between two or more wave trains, whereas rapid development of sea index allows for the effects of rapid changes in sea state on boats and its capsizing. The preliminary thresholds of all these indices were determined by studying the wave model output for all the high wave alert conditions for an entire year. Accordingly, the thresholds for steepness index, crossing sea index, and rapid development of sea index were determined to be 0.8, 0.65, and 0.2 respectively. The preliminary thresholds for these indices will be refined periodically based on the inputs from a strong feedback mechanism, over a while for the region under consideration. A single index named BSI that takes into consideration the combined effects of all the three indices, and their severity is proposed and introduced. This simplifies and combines the whole information regarding the dangerous sea states, thus helping in the identification of regions with a high probability of boat capsizing. The validity of the BSI is established by carrying out verification with real-life accident case studies. The accident cases simulated and analyzed in this work for the verification studies covered the southern and northern parts of the study domain. It can be concluded from the verification studies that the regions where capsizing occurred had $BSI > 0$, and were found to be dangerous as per the advisory system. Thus, the results of the present study are encouraging and give us confidence in using the forecast and advisory system developed, verified, and operationalised at ESSO-INCOIS. The advisories in these high probability capsizing regions are tailored as per the beam size of the vessels, eventually making it boat-specific. This advisory services system is semi-operational and is expected to be fully operational over some time with the help of a robust feedback mechanism. To further reduce the boat accidents, the operators must be thoroughly trained in the safety measures that should be taken into consideration while on the boat, and measures to be taken in the case, the boat capsizes, to minimise the losses.

This is a first-of-its-kind impact-based forecast service system, globally, that pre-warns the sea-farers regarding the probability, location, and time of boat capsizing. Hence, there was no role-model available to follow in any of the aspects such as design, defining the thresholds and advisory format, etc. while it was being developed. However, once the complete system is operational, we will start receiving feedback directly or through our NGO/institutional collaborative partners. INCOIS, being an operational oceanographic forecasting and service centre right from its inception,

routinely analyzes such feedback for the existing operational services and refines the forecasting/advisory system to suit the reality. Similarly, we plan to analyze the feedback for SVAS and refine the system by all possible methods including re-defining the thresholds used in the BSI. This will eventually lead to an improved small vessel advisory and forecast service system for India in the coming years.

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No potential conflict of interest was reported by the author(s).

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